Reinforcement Learning Laboratory

Lab Assignment 3: GridWorld – Policy Evaluation and Value Iteration

Course Outcomes: CO2, CO3, CO4

Objective

Formulate a GridWorld environment as a Reinforcement Learning problem, implement Policy Evaluation and Value Iteration, and compare different policies.

Instructions

- Study the GridWorld environment and identify states, actions, rewards, and terminal states.
- Implement Policy Evaluation for a given policy.
- Implement Value Iteration to determine the optimal policy.
- Compare paths generated by random and optimal policies.
- Record observations and answer all analysis questions.

Task 1: Formulating the GridWorld Environment

RL Component	Description
States	S1, S2, S3, S4, S5
Actions	UP, DOWN, LEFT, RIGHT
Reward Structure	−1 for every movement, including invalid movement; 0 at terminal state S5
Terminal State(s)	S5
Goal of the Agent	Reach S5 from the starting state S1 using the minimum number of steps

Task 2: Policy Evaluation

Iteration	Maximum Change (Δ)	Convergence Status
1	*1.0*	*Not converged*
2	*0.9*	*Not converged*
3	*0.0*	*Converged*
4	*0.0*	*Converged*
5	*0.0*	*Converged*

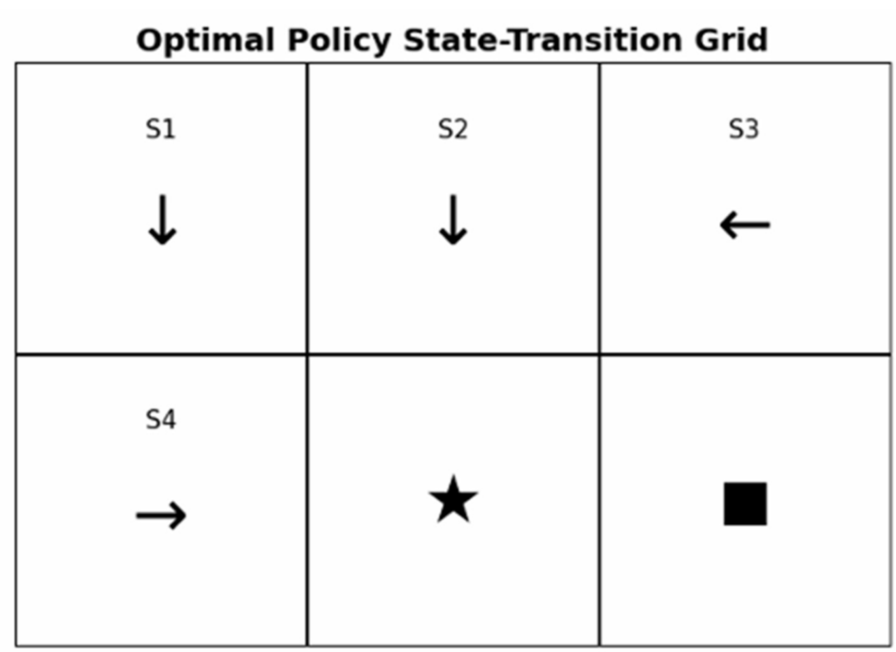
Task 3: Value Iteration and Optimal Policy Generation

State	Optimal Action	State Value
S1	*DOWN↓*	*−1.9*
S2	*DOWN↓*	*−1.0*
S3	*LEFT ←*	*−1.9*
S4	*RIGHT →*	*−1.0*
S5	*Terminal*	*0.0*

Task 4: Optimal Path Analysis

Random Policy	**9**	**−9**	**Yes**
Evaluated Policy	**2**	**−2**	**Yes**
Optimal Policy	**2**	**−2**	**Yes**

Optimal Policy Representation of the GridWorld using Directional Arrows:



Analysis Questions

1. What is a state in GridWorld?

Answer:
A **state** represents the current position or situation of the agent in the GridWorld environment. In this experiment, the states are S1, S2, S3, S4 and S5.

Example
If the agent is currently at S2:

Current State = S2

The agent then chooses an action based on that state.

2. What actions can the agent perform?

Answer:
The agent can perform four actions:

- UP
- DOWN
- LEFT
- RIGHT

These actions determine the direction in which the agent attempts to move within the GridWorld.

3. What is the purpose of Policy Evaluation?

Answer
The purpose of **Policy Evaluation** is to calculate the value of each state when the agent follows a given policy.

In this experiment, the given policy is:

S1 → RIGHT

S2 → DOWN

S3 → LEFT

S4 → RIGHT

Policy Evaluation repeatedly updates the state values until they converge.

4. How does Value Iteration differ from Policy Evaluation?

Answer
Policy Evaluation evaluates a fixed policy, whereas **Value Iteration** considers the possible actions at each state and selects the action that provides the highest expected return.

Policy Evaluation

↓

Given policy

↓

Calculate state values

Whereas:

Value Iteration

↓

Consider possible actions

↓

Calculate their values

↓

Choose maximum

↓

Optimal policy

Policy Evaluation tells us how good a given policy is, while Value Iteration finds the best policy.

5. What is the Bellman Optimality Equation?

Answer
The Bellman Optimality Equation used in this experiment is:

$$V^*(s) = \max_a [R(s,a) + \gamma V^*(s')]$$

where:

- **V*(s)** = optimal value of the current state
- **R(s,a)** = immediate reward
- **γ** = discount factor
- **V*(s')** = optimal value of the next state
- **max** = choose the action with the highest value

The equation says: **Choose the action that gives the best combination of immediate reward and future reward.**

6. Which policy reached the goal using the fewest steps?

Answer
Both the **Evaluated Policy** and the **Optimal Policy** reached the goal using the fewest steps.

They both required:

- 2 steps
- The Random Policy required:
- 9 steps

The two shortest paths are:

- Evaluated: S1 → S2 → S5
- Optimal: S1 → S4 → S5

Both have:

- Path Length = 2
- Total Reward = -2

7. Why is the optimal policy superior to a random policy?

Answer
The optimal policy is superior because it chooses actions based on the estimated values of future states and therefore avoids unnecessary movements.

In this experiment:

Policy	Steps	Reward
Random	9	-9

Policy	Steps	Reward
Optimal	2	-2

Therefore, the optimal policy:

- reaches the goal faster,
- avoids unnecessary movements,
- obtains a better total reward,
- and follows a goal-directed path.

8. How would obstacles affect the optimal path?

Answer

Obstacles would block some possible movements and prevent the agent from directly reaching certain states. The agent would then have to find an alternative route around the obstacles. This could increase the number of steps required to reach the goal and would change the optimal state values and the resulting policy.

Conclusion

Policy Evaluation was used to determine the value of each state under a fixed policy and to observe how these values converged over successive iterations. In the implemented GridWorld, the state values converged by the third iteration.

Value Iteration considered all available actions at each state to determine the optimal state values and the corresponding optimal policy.

The Random Policy required 9 steps and obtained a total cumulative reward of −9, whereas both the Evaluated Policy and the Optimal Policy reached the goal in 2 steps with a total cumulative reward of −2.

These results demonstrate that goal-directed policies provide more efficient behavior than random action selection. The shortest-path comparison further shows that reaching the goal in fewer steps results in a better cumulative reward for the given GridWorld environment.

Overall, this experiment illustrates how Policy Evaluation measures the quality of a fixed policy, while Value Iteration builds on that process to discover the optimal policy — and how the presence of obstacles would further influence these state values and the resulting optimal path.